



ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

THE ECOLOGY OF EUCALYPTUS

ANDRÉ DA GLÓRIA SANTIAGO

Supervised by : Elisabeth Davin-Mortier and Jérôme Baudry

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ABSTRACT

This project aims to do a corpus analysis of the discourse around eucalyptus within the French-speaking world. The goal is to perform an exploratory analysis of the field, and understand who are the actors and how do they shape the discourse. To do so, we downloaded a corpus of 17'191 total documents spanning from 1783 to 1920 to which we applied a topic model. Based on the analysis performed on the documents' metadata and the results of the topic modeling, we then focused on three main case studies: the documents published in Lyon and their links with advertisements, the documents published in Madagascar and their ties with the French colonial administration and finally an analysis on the topic of drainage of the soil and the fight against malaria. This methodology offers a new way to explore the social and political ramifications of the eucalyptus tree, as well as identifying new potential cases for more in-depth qualitative research.

1. INTRODUCTION

Eucalyptus is a tree that, while nowadays common in the Mediterranean landscape, has only been around Europe for less than 200 years. Originally from Australia, it was first discovered by Europeans in the late 18th century, following the maritime expeditions of James Cook, through which some samples of the plant were brought over to Europe. It was based on these that the botanist Charles Louis L'Héritier gave the plant its current name: "Eucalyptus". However, it is not until the second half of the 19th century that the tree definitely entered the Mediterranean landscape, notably through the works of Prosper Ramel, who, after being key in the import of the tree to Europe, started large plantations of eucalyptus in France and Algeria¹. These new plantations, in turn, fuel a discourse about this new exotic tree, which had many virtues and qualities; both industrial (strong wood for construction) and domestic (folk medicine) in nature.

Current research is lacking in the analysis of the discourse around the importation and the spread of eucalyptus throughout the French-speaking world. We can cite the work of Robin W. Doughty², tracing the history of eucalyptus and its usages, however more from an anglophone perspective. We can also cite the work of Elisabeth Davin-Mortier, who focuses more on Europe, the Middle-East and more generally on the Mediterranean basin, looking at the links between the plant and its role in European colonization. Thus, the aim of this project is to address this gap, via a quantitative approach, by analyzing more than 17'000 different documents, published between 1783 and 1920. We chose to limit the corpus at the end of the first world war, as it represents a key event impacting the regions studied, at a time where we can already find large eucalyptus plantations around the Mediterranean. So, in this project, we will try to understand how and through which actors the discourse around eucalyptus is shaped. More precisely, understand in which contexts the tree is mentioned and with what kind of properties it is associated in the documents. The analysis will be pursued both through time and space, to see how the discourse evolves, and from where it originates.

¹Elisabeth Davin-Mortier. *L'eucalyptus : un arbre aux ramifications environnementales, sociales et politiques en Méditerranée (XIXe-XXe siècles)*. 2026, p. 2

²Robin W. Doughty. *The eucalyptus : a natural and commercial history of the gum tree*. Center books in natural history. Baltimore: Johns Hopkins University Press, 2000. ISBN: 0-8018-6231-0

2. METHODS

2.1 CREATING THE CORPUS

2.1.1 DOWNLOADING THE FILES

In order to analyze the evolution of the discourse about eucalyptus, we need a large number of relevant documents. Thus, a corpus was formed through Gallica's API (the online platform of France's national library). Due to the nature of the API, we first downloaded all the metadata of the documents that mentioned the word "eucalyptus" in its text, resulting in a total of 58'157 different documents ¹. However, due to time and API restrictions, we decided to download only a portion of the potential documents. Thus, we kept only the ones having in their metadata either a specification of its author or its publisher and mentioning the year they were published in. Figure 2.1 shows the yearly distribution of the full corpus, compared with the filtered one. We do see that general trends prevail, while certain spikes are smoothed out. Looking more in-depth, we see that most of the filtered out documents were published in Paris, and the same for the spike in 1900. Moreover, most of the removed documents belonged to the press, meaning that, should an event happen, it would probably also be captured by another journal within the kept documents. While this means that the final corpus might not capture all local phenomena, the loss seemed reasonable, as we expected a heavier usage of the author and publisher fields. We do agree, however, that the filtering is not perfect, and looking back a random sampling would have been more relevant. Indeed, some issues arise because of the inconsistency of Gallica's metadata concerning the press, as some journals have themselves as their own publisher, whereas some do not, thus being removed from the pool of potential documents. Since this metadata decision from Gallica seems to be arbitrary and was not caught beforehand, it does present a bias in the corpus. However, given the structure of the API and time limitations that did not allow for a reevaluation of the filtering, we think the final corpus is still relevant, and the wider one (without full document information) will be used periodically if needed.

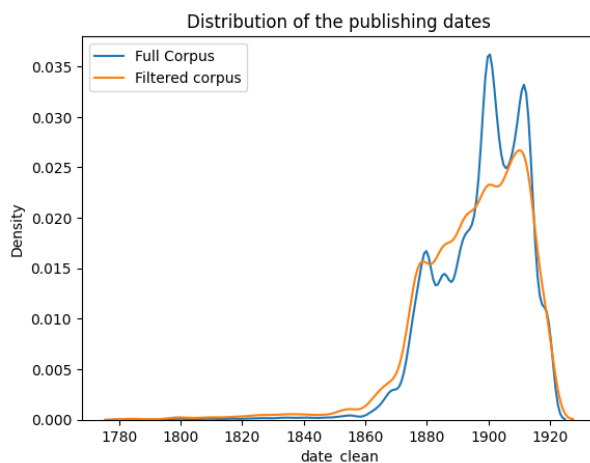


FIGURE 2.1
Histogram depicting the distribution of documents in the two corpus per year.

¹The script was run in early March. Due to more documents being added, the number may have increased at the time of reading.

In the end, the final corpus contains a total of 17'191 documents, ranging from 1783 to 1920. Each document possesses a title, a date of publication, either a stated author or publisher and its type (monograph or press). Moreover, the author field contains the birth and death of the authors, whereas the publisher field (even if there is no publisher stated) contains the document's place of publication. Finally, each document has its text file, directly downloaded via Gallica's API, thus relying on their own OCR model. This proves to be a slight problem, as the model used cannot be considered as "state-of-the-art" for today's standards, and a lot of errors appear, not only from misspellings but also from text structure, as their model struggles with press layout, often separating two paragraphs of the same article that should be together.

2.1.2 ENRICHING THE METADATA

While having assembled a corpus of documents related to eucalyptus, we found that some metadata fields were either not precise enough, too precise or simply lacking. Thus, based on the existing metadata, we decided to create three new fields for our analysis: the author type (person or entity), the typology of the document (monograph, generalist press, specialized press, official press or directory) and the region where the document was published.

Concerning the author type, the general principle was to use the Spacy's entity tagging to check if the author field was a person². Then, should the model not detect the name of a person, the author field was matched against a list of french names. In the end, if both tests did not detect the name of a person, it was considered as an entity (for example a private journal, an association or a state). A manual test of a sample of 100 documents was conducted, showing no errors in this field.

A similar process was conducted for the type of publication. Gallica's own type field only differentiated between "monograph", "printed serial" and "directory". Keeping "monograph" and "directory" as its own entries, the challenge was differentiating between the different types of "printed serial", such as "generalist press", "official press" and "specialized press". We felt that this distinction was key for the analysis, as different types of journals have different purposes and editorial objectives. Indeed, a medicinal journal will not have the same approach as a daily newspaper, especially about the subject of eucalyptus. To differentiate between these categories, we tried to both detect keywords in the titles and use a list of known generalist newspapers. Thus, words such as "médecin, académie" were considered as "specialized", "express, eglise, gazette, dépêche" as "generalist" and "budget, procès-verbaux, officiel" as official press. A continuous process of verification of the results produced by the algorithm was used to refine the results. In the end, manual verification over 100 samples only showed 1% of error.

Finally, we regrouped the places of publication into more general regions, based on geographic and cultural factors. Thus, we have 7 "regions" in the corpus: "Île de France/Paris", "South of France", "Metropolitan France", "North Africa", "Madagascar", "Europe" and "Other". While separating Île de France and the rest of the country is reasonable given the cultural status of the capital, we need to explain the separation between "South of France" and the rest of "Metropolitan France". When discussing eucalyptus, this distinction is pertinent, as it was in the south, notably in cities such as "Montpellier" or "Toulon"³, that eucalyptus were planted. We decided to use the current administrative regions' borders to divide between North and South, while fully acknowledging that they are anachronistic compared to the period studied. This decision was done so that the delimitation was easier to program. In the end, the following regions were considered as "South": Nouvelle-Aquitaine, Occitanie, Auvergne-Rhône-Alpes, Provence-Alpes-Côte d'Azur and Corse. While it may not exactly align with a traditional North/South divide, it matches well enough with the regions influenced by large plantations of the tree. As with the other fields, a manual verification was conducted over 100 samples and we only found 1% of error.

²Ines Montani et al. *explosion/spaCy: v3.7.2: Fixes for APIs and requirements*. 16th Oct. 2023. DOI: 10.5281/ZENODO.1212303. URL: <https://zenodo.org/doi/10.5281/zenodo.1212303> (visited on 3rd June 2026)

³Doughty, *The eucalyptus : a natural and commercial history of the gum tree*, p. 45

2.2 TOPIC MODELING

The next step in the analysis of the discourse was the implementation of a topic model, which aimed to capture the different ways eucalyptus are talked about. To do this, the BERTopic model was used, as it is currently one of the best performing open models for such an endeavor⁴. However, we did not simply feed every document to the machine and found the corresponding topic. Indeed, the document typologies discussed in the section above have different structures, some of which would not work well for our purpose if done in such a way. For example, the generalist press might (and generally will) only have a single article about the tree within its pages. As such, capturing the topic of the whole document would not capture the information that we need. That is why it was decided to use as input every sentence where the word "eucalyptus" appeared on. There are several reasons justifying this choice. First, it allows for better capturing the different manners of discourse within a document itself. A document might conceivably talk about the origins of the plant at one point and then about its medicinal properties. A separation by sentence allows to capture both. Second, it allows to use a similar procedure for all document types, as the context window is small enough to avoid taking in irrelevant information, while seemingly wide enough to contextualize it. Finally, after performing topic modeling across different context sizes, it was found that heuristically it worked best (more sentences includes often different articles, while paragraphs are often badly detected by the OCR process, especially for press articles). However, this method might skew some distributions towards documents where the word "eucalyptus" appears very often. To mitigate this, the graphs presented throughout this paper, unless specified, have been normalized so that every document has the same weight.

The model having found 135 different topics, we decided it was best to group them into shared themes, or meta-topics. We chose to represent 10 different categories: "Geography", "Eucalyptus and Botanical Properties", "Fauna and Flora", "Malaria and Drainage", "Diseases", "Medicine and Medication", "Chemical Properties", "Industry and Manufacturing", "Institutions and Transport", "Descriptions" and "Others". An 11th "Errors" meta-topic was also created, which detected OCR errors ($\sim 7\%$ of all contexts). These labels were manually curated and were the fruit of a discussion between myself and this paper's supervisor. While we mainly had a bottom-up approach, regrouping topics by shared theme, we also decided to include some that we were looking for in the corpus, such as "Malaria and Drainage", as we knew that it was a key reason for eucalyptus plantations, thus not including "malaria" in the "diseases" meta-topic, and grouping it with "drainage" instead. We also decided to merge "Eucalyptus" and "Botanical Properties" as we figured that given the context, these properties were directly referencing the tree, as well as merging "Institution" and "Transport" as instead of merging the few "transport" topics into "Others", they were still relevant when linked with "Institution". However, we kept the rest of "Fauna and Flora" separated from eucalyptus, as in the context of studying the tree, it made sense to distinguish what is related to eucalyptus and what is related to the general fauna and flora. Finally, it was also decided to keep "Diseases", "Medicine and Medication" and "Chemical Properties" separated instead of a larger "Medicine" meta-topic, as they are different parts of a larger theme, that one could expect to appear in different types of documents.

Lastly, we used other sporadic methods in the following analysis. First, we have taken into account temporality, and used built-in methods in BERTopic to analyze the distribution of our topics through time. However, it takes the whole corpus into account, so when needed, a similar method was done, showing the the number of contexts associated with a topic though time. Finally, a co-occurrence graph was also constructed, with topics as nodes and the number of documents in which they appear simultaneously as weighted edges. This will also be used when needed to show the relationship between two topics.

⁴Maarten Grootendorst. 'BERTopic: Neural topic modeling with a class-based TF-IDF procedure'. In: (2022)

3. RESULTS

3.1 GENERAL OVERVIEW

To understand how eucalyptus is talked about, we first need to understand who is speaking about it. As such, we can first divide our dataset between two large groups of actors: persons and entities. In Figure 3.1 we see the distribution of these two categories per year. While before 1860 the number of authors is low, documents about eucalyptus are mostly being published by authors. However, the number of "entities", rises after 1860. This is related to the general rise of articles talking about this new plant, which is thus mostly driven by entities.

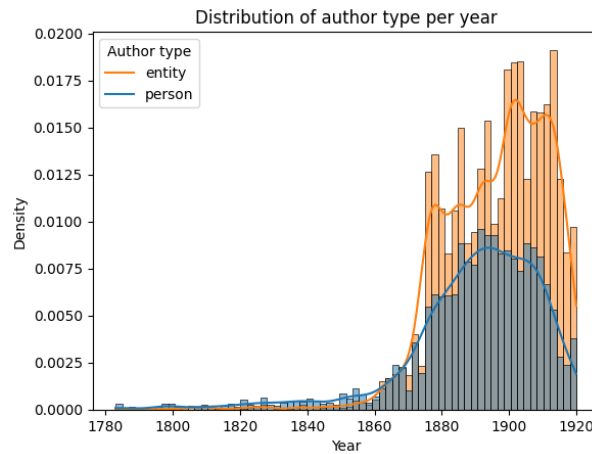


FIGURE 3.1

Histogram depicting the distribution of documents per author type and per year.

However, we can distinguish more than one type of entity that might be interested in publishing about eucalyptus. As such, we can divide these two categories (person and entity), into a typology of documents: monographs (which will mostly be published by "persons"), generalist press, specialized press, official press and directories. The distribution of authors and document type is shown in Figure 3.2, and it clearly follows what we would expect: monographs are published by "persons" and the rest by "entities".

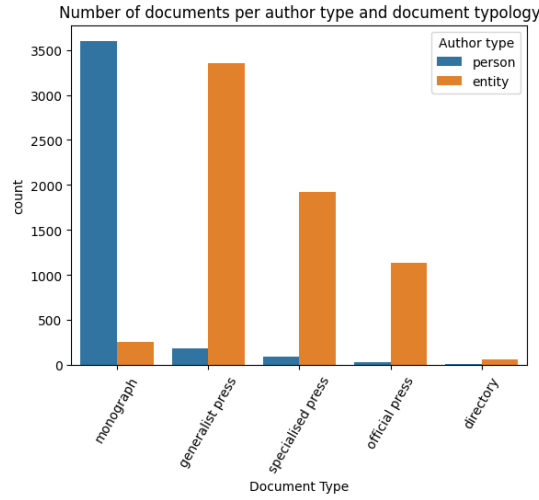


FIGURE 3.2

Bar plot displaying the number of documents per author and document type

It is then useful to look at the distributions of these different types of actors through time, to see if we discover certain patterns. Figure 3.3 does show that the rise in the amount of documents is driven by the generalist press, while monographs and the specialized press follow a similar trajectory. While this could be an artifact of Gallica’s corpus, we see in Figure 3.4 that it should not be the case. While there is a gradual increase in the number of documents in Gallica’s database until 1914, the sudden rise in documents related to eucalyptus in the 1870s is a different phenomenon. Moreover, the frequent ups and downs (i.e. the drop in the 1880s) in the amount of generalist documents compared to the other types do show a more fleeting usage of the word, possibly driven in the corpus by certain advertisement campaigns or punctual discussions on government policies. Finally, Figure 3.3 does show an uptick in the official press starting from 1900, which is, as is further developed in Section 3.3, mostly related to the colonization of Madagascar.

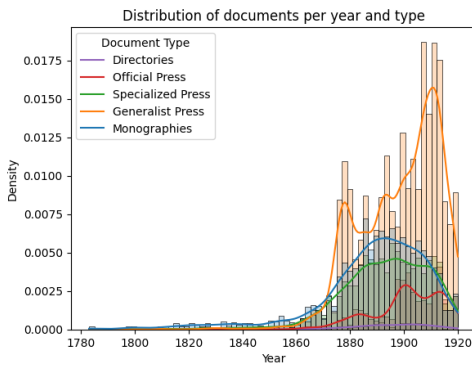


FIGURE 3.3

Histogram displaying the distribution of document types per year.

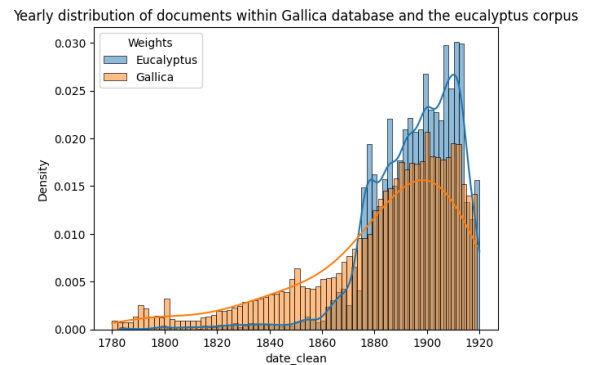


FIGURE 3.4

Histogram comparing the yearly distributions of the eucalyptus corpus and the overall Gallica corpus.

We can also look at the distribution of documents per place of publication as in Figure 3.5. While the presence of Paris is expected, we do see a rise in documents published in the south of France starting from 1900, driven by the city of Lyon, which will be further developed in Section 3.2. Moreover, the shape of the curve of documents published in Madagascar is quite similar to the official press, showing, as mentioned previously, the relationship between the two. The change in the 1860-70s is also visible,

with an uptick of documents published in Paris, South of France and North Africa, while in the rest of Metropolitan France the rise is more steady. It probably comes from the fact that it is not a region of plantation and thus the number of documents does not share the same peak.

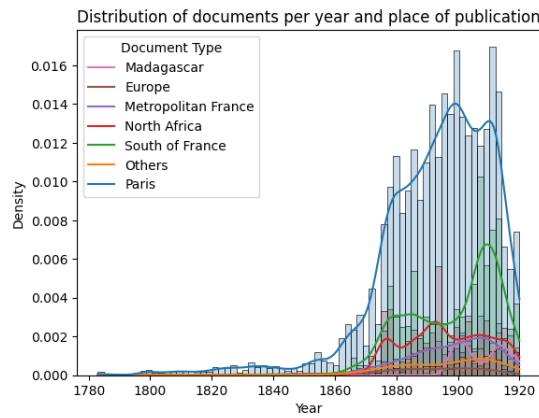


FIGURE 3.5

Histogram displaying the yearly distribution of the publishing places of documents.

Finally, to understand how eucalyptus is spoken about in the corpus, we can use topic modeling, as described in Section 2.2. Table 3.6 gives an overview of a few of the most relevant topics in the documents, while a full account can be found in Appendix A. The biggest one is topic 0, which talks about the place of origin of the tree. Given that it is a new plant in Europe at the time of study, the emphasis on Australia is not surprising. The second biggest topic however is about Algeria and North Africa. While it may be unexpected to see it so high, the tree was heavily planted in the region after 1970. The fact that "Trottier", a key figure in the implementation of eucalyptus plantations in Algeria, appears in the topic shows the relationship between the region and the tree and why it is a key topic in the corpus. Finally, topics 3 and 9 show a key reason for the plantation of eucalyptus. Indeed, the tree was used as an answer to drain marshes, both to fight against malaria and to transform them into arable land, more suitable for European settlers ¹. In the end, the model found 135 different topics, with a heavy tailed distribution (Figure 3.7), emphasizing the importance of the larger topics, but with smaller ones still having a good number of contexts.

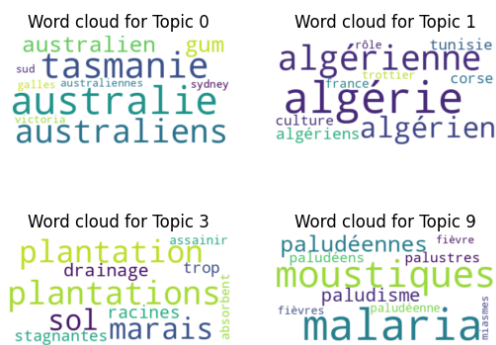


FIGURE 3.6

Word clouds displaying the words most associated with topics 0, 1, 3 and 9.

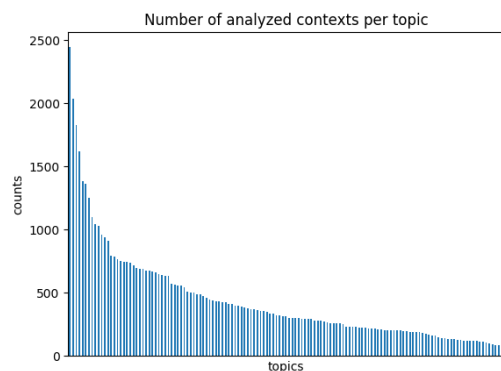


FIGURE 3.7

Histogram displaying the distribution of the number of contexts per topic.

¹Davin-Mortier, *L'eucalyptus : un arbre aux ramifications environnementales, sociales et politiques en Méditerranée (XIXe-XXe siècles)*, p. 6

Given their amount, we can group them into meta-topics, and analyze their distribution within the typology previously discussed. The result is presented on Figure 3.8, showing the distribution of each meta-topic within each category. The results do show a prevalence for Institutional topics within the official press, more scholarly topics (medicine and botanical and historically related) in monographs and specialized press and a bigger amount of "Others" in the generalist press. We can also look more at the way the medicine-related topics are distributed. We really do see this separation between monographs and the specialized press against the generalist press, whereas the first group do have a good amount of contexts related to the chemical properties of the plant and its medicinal and medication uses, the generalist press only covers medicine and medication (and in a smaller amount), probably related to advertisements of such products, as will be seen further in the analysis.

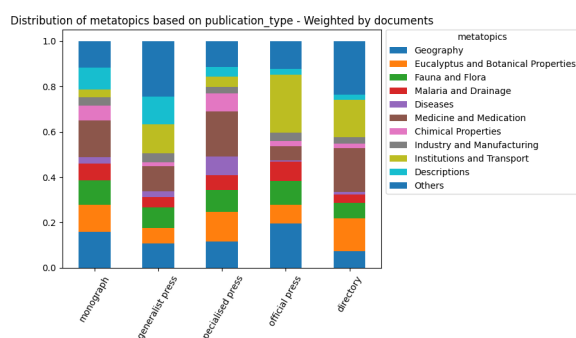


FIGURE 3.8

Bar plot displaying the distribution of meta-topics within document types.

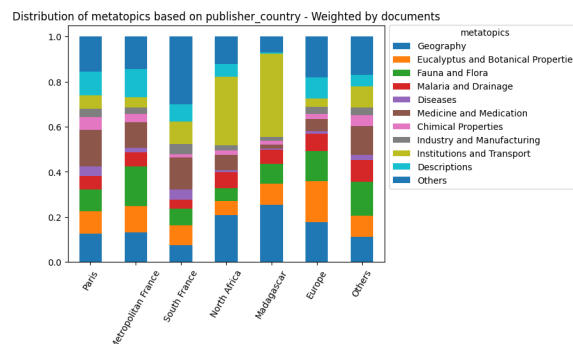


FIGURE 3.9

Bar plot displaying the distribution of meta-topics within publishing regions.

We can also look at the meta-topic distribution within place of publication (Figure 3.9). We see similar distributions between Paris, Metropolitan France and Others, while topics in North Africa and Madagascar are clearly skewed towards Institutional ones. It is interesting to note that both colonized regions appear with similar distributions, as it shows that the tree does appear in similar way in the context of colonization, notably through institutional means. At last, we can notice that the topic "Malaria and Drainage" does appear consistently around 5% in each region. Historically one of the main drivers of eucalyptus plantation, while it does not seem to be the main driver of discussion in published documents, it is still consistently in the same proportions in each region, showing that it remains a key reason for plantation and thus in the discussion around the tree. The fact that it appears only around 5% is thus probably linked with the number of generalist press, which will likely talk less about this subject.

After this overview of the results, three case studies will be analyzed in depth in this section, based on the interesting results from the overall data. First, we will look at the rise in documents in Lyon starting from the 20th century, then at the publications from Madagascar in the same time period, and finally at the evolution of the documents related to the drainage of marshes and the fight against malaria.

3.2 LYON BETWEEN 1900 AND 1920

In order to understand the importance of eucalyptus in Lyon during this time period, we first need to better understand what makes it a good case study. Looking at the distribution of documents per year and place of publication in Figure 3.5, we see a clear rise in the documents published in the south of France starting from 1900. When investigating closer, we see that after 1900, 67% of documents published in the south of France were from Lyon, whereas the proportion before 1900 is only of 20%. This change shows that a phenomenon happened within the documents published in the region.

We can then look at which documents influence this rise. As Figure 3.11 shows, it is clearly pushed by a rise in the generalist press, led by the journal "Le Progrès", which accounts for 1187 of the total 1458 documents published in Lyon during this time frame. We can discard the hypothesis of the implementation of a new journal, as, according to Gallica, "Le Progrès"'s first publication was in 1859 and the journal has been published daily since then. Thus, as eucalyptus appearances only start from 1900, this is clearly something new happening in this particular journal.

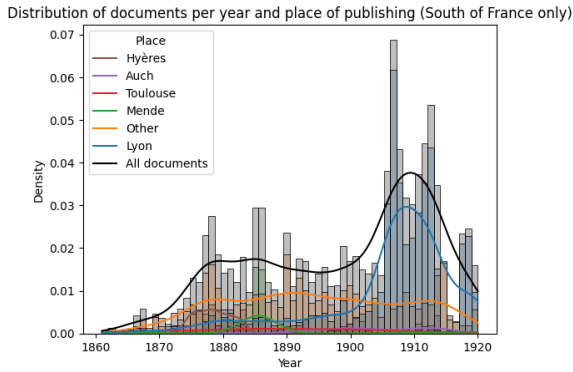


FIGURE 3.10

Histogram displaying the yearly distribution of documents per cities in Southern France.

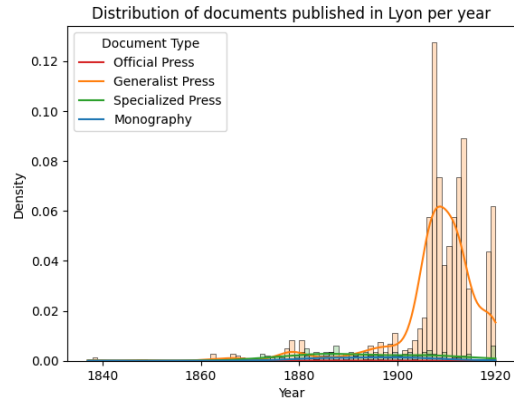


FIGURE 3.11

Histogram displaying the yearly distribution of documents published in Lyon per document type.

To understand what it is talking about, we can look at the topic distribution within the documents published in Lyon, as shown in Figure 3.12. We clearly see that the meta-topic "Others" is in 40% of documents published in the city. By looking more in-depth at the topic distribution within "Others", we find that 50% of those is the topic 46. It speaks about pharmaceutical entities, such as "elixir" or "vermifuge", while also mentioning some units of measure as "litre" or "kilo". Looking at it without context, the words probably indicate a topic linked with a specific pharmaceutical advert. And indeed, we do find such and advert in several issues of "Le Progrès", as seen in Figure 3.13.

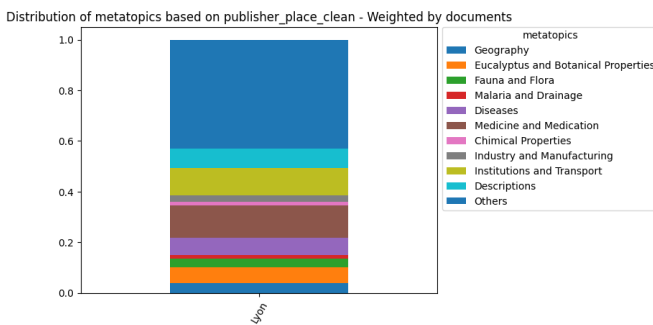


FIGURE 3.12

Bar plot displaying the distribution of meta-topics within Lyon.



FIGURE 3.13

Advert for the "Grande Pharmacie Lyonnaise"²

To see if such topics are indeed what drives this spike in Lyon, we can look at the topics over time. Topics

²"Le Progrès : journal de Lyon, politique, quotidien". In: (4th May 1906). URL: <http://catalogue.bnf.fr/ark:/12148/cb32843715j> (visited on 4th Feb. 2024)

46 and 88 represent respectively 25% and 8% of all topics in Lyon and both are linked with advertisement. The first, as seen in Figure 3.13 with the "Grande Pharmacie Lyonnaise", and the second with an advert of an eucalyptus and honey based medicament against tuberculosis ("War against Tuberculosis")³, which explains why a word such as "war" is linked with more medical terms such as "tuberculosis" or "pectoral". By looking at their frequency over time (Figure 3.14, we do see that they are highly focused in certain time-frames: 1908-1913 for topic 46 and 1904-1906 for topic 88. These match with the rise of documents mentioning "eucalyptus" published in Lyon, thus showing that advertisements are a key medium through which eucalyptus is talked about. Moreover, it is also interesting to note the environment in which the tree is used within these advertisements. Both are linked with the pharmaceutical and medicine industry, thus already showing its place as a plant with medicinal virtues in the collective mindset.

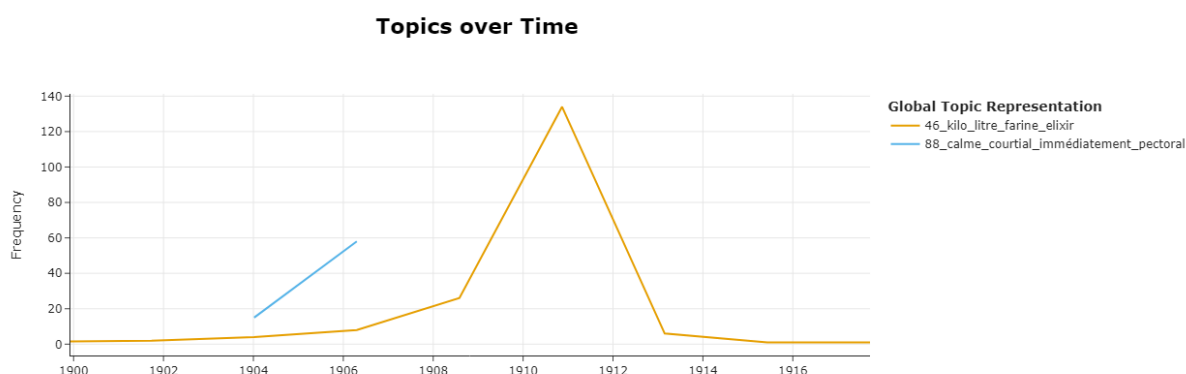


FIGURE 3.14

Line plot displaying the yearly frequency of topics 46 and 88 in the corpus.

To conclude, while Lyon is a clear case study of publicity in relation with eucalyptus, this phenomenon appears throughout the corpus, and is often captured in specific "advertisement" topics. These are often local to specific cities and capture advertisements that appear throughout several documents. Topics 46 and 88 for example, appear overwhelmingly in Lyon only, with respectively 78% and 62% of the documents related to those topics coming from there. Thus, other similar "advertisement" topics can be found, such as topic 72 with 52% of its documents published from "Mende" (who only represents 0.6% of the total number of documents). Such appearances of advertisements should not be overlooked when analyzing the discourse around eucalyptus, as it shows us what is the appeal of the tree. Indeed, advertisements aim to sell products and as such will use (or try to build) a shared imagery on what the plant is and what its qualities are. Thus, as advertisements are mostly linked with medicinal properties and usages, these are clues to understand what the general population thinks, or is invited to think, of the tree.

3.3 MADAGASCAR AND THE OFFICIAL PRESS

While Lyon showed the importance of the adverts in the number of documents related to eucalyptus, Madagascar shows the importance of the official instances, and how this plant seems to be used as a tool for land planning. Indeed, it was in 1896 that French troops invaded and took control of the island⁴, and

³'Le Progrès : journal de Lyon, politique, quotidien'. In: (22nd Dec. 1907). URL: <http://catalogue.bnf.fr/ark:/12148/cb32843715j>

⁴Samuel F. Sanchez. 'De l'État royal à l'État colonial à Madagascar : Ruptures et continuités des politiques économiques (fin XIXe-début XXe siècle)'. In: *Madagascar, d'une crise l'autre : ruptures et continuité*, Paris, Karthala-IRD.. Ed. by Razafindrakoto M, Roubaud F and Wachsberger J.-M. 2018. URL: <https://shs.hal.science/halshs-02079018>, p. 79

this colonization process is seen in the corpus, as the first instance of the "Journal officiel de Madagascar et dépendances" in the corpus dates from 1897, and until 1920 it appears 459 times (around 20 instances per year for a journal published around once or twice a week, based on Gallica's data). Moreover, this journal accounts for 86% of all documents published in Madagascar within the corpus, showing both the importance of the official documents in relation with eucalyptus, and seemingly the lack of a generalist francophone press at the early stages of the colonization process.

Having established that eucalyptus were talked about in Madagascar's official documents, it is important to learn how it was talked about and for which purposes it was used. We can thus look at the distribution of meta-topics within the documents published in the island (Figure 3.15). It shows us that close to 40% of the topics mentioned in the area belong to "Institutions and Transport", a bit more than 20% to "Geography", while meta-topics related to technical and scientific skill ("Medicine", "Chemical Properties" and "Industry and Manufacturing") are almost negligible.

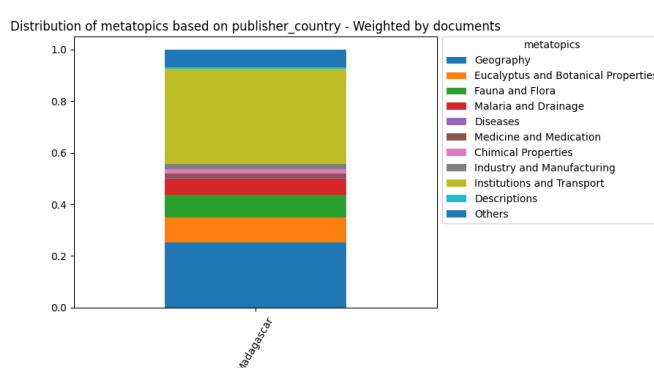


FIGURE 3.15
Bar plot displaying the distribution of meta-topics within Madagascar.

Based on the previous distribution, it is important to focus on the topics belonging to "Geography" and "Institutions". Looking at both, topic 43 represents 80% of all the "Geography" meta-topic (and ~30% in total), while topic 80 represents close to 75% of all "Institutions" meta-topic (~23% in total). This shows that the tree appears in rather uniform contexts within the island's publications. It is interesting to note that the third biggest topic is topic 3. While its distribution within Madagascar and the rest of the corpus is not that different (~5%), it shows that drainage is still one of the main roles fulfilled by eucalyptus in Madagascar. Moreover, there is close to no document co-occurrence between topic 3 and topics 43 and 80 (only 2 and 3 documents respectively), showing that the issue of soil drainage is spoken about in different contexts than issues related to the institutions (which often co-occurs with the geography topic).

Looking at the words inside the biggest topics, number 43 represents Madagascar ("Fianarantsoa", "Malgaches"), its regional administration ("district", "canton") and administrative terms, probably related with its colonization history ("réquisition", "né(e)"). As such, it does make sense that it appears as the biggest topic in the island's publications. Topic 80 on the other hand has words in the lexical field of decision making and administrative settings ("déclaré", "demandé", "propriétaire"). Knowing that most of the documents published on the island are from the official press, these words make sense. However, it is interesting to note other words in the topic, such as "patates" or "manioc", showing that eucalyptus appears in the same context as food production. To better understand it, we can look at examples of documents in which topic 80 appears, such as in Figure 3.16. This document describes the requisition of a property close to Tamatave, which is then received by a local administrator. Eucalyptus are thus present in the description of the property's terrain, as well as the above-mentioned "patates" and "manioc". This example, far from being isolated, shows that the tree is already quite established in the island and is

discussed in the same way as other food related plants.

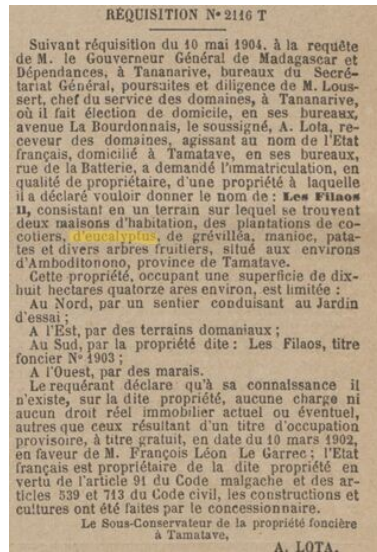


FIGURE 3.16
Requisition of a land parcel close to Tamatave⁵

By looking at the amount of contexts by year, related to the three topics discussed above (3, 43 and 80) by year (Figure 3.17), we see that topic 3 is quite infrequent but relatively stable. However, topics 43 and 80 clearly rise starting from 1910, but then drop at start of the war. This probably shows one of the reasons why eucalyptus are being planted in Madagascar, but also the amount of properties which already have the tree in their descriptions.

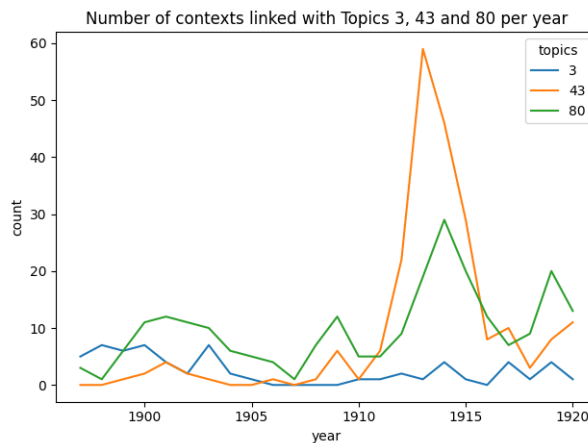


FIGURE 3.17
Line plot displaying the yearly number of contexts in Madagascar linked with topics 3, 43 and 80.

In conclusion, we see a region that immediately after the French conquest starts producing literature that mentions eucalyptus. These documents are heavily linked with institutions, showing that there is not yet a solid French presence on the island, as there is not really any generalist press for local settlers, as can be seen in Algeria. Here, most documents concern administrative regions and the requisition of properties

⁵Madagascar. 'Journal officiel de Madagascar et dépendances'. In: 20.909 (8th June 1904). URL: <https://gallica.bnf.fr/ark:/12148/bpt6k6485848h>

by the settlers. The tree is here often mentioned alongside plants related to the food industry in terrain descriptions, showing that it is already present in the island. Moreover, the overwhelming presence of institutional topics, as well as the topic of malaria and drainage also tends to show that it participates in certain landscaping efforts either by the state or private actors.

3.4 DRAINAGE AND THE FIGHT AGAINST MALARIA

While the two previous case studies had a data-driven approach, in the sense that the data showed patterns linking Lyon with advertisements and Madagascar with institutions and colonization, this one aims to analyze a topic that we know is key in respect to the usage of the plant. Indeed, one of the main purposes of eucalyptus plantations during this time period is the sanitation of marshes, notably in the context of the fight against malaria⁶. Indeed, the roots of the tree quickly dried out marshes and swamps, while, at the same time, seemingly being an effective remedy against malaria outbreaks⁷. It is reportedly during a serious outbreak between 1867 and 1876 in Algeria, that plans of eucalyptus reforestation were carried⁸. Thus, knowing that this was an important reason for eucalyptus plantation, it is interesting to understand how the discourse around the tree in respect with the topic of drainage is shaped.

In our data, we have a meta-topic "Malaria and Drainage" (comprised of topics 3, 9, 29 and 104) that we can use to analyze this issue. First, we see that across the dataset, around 5.5% of the contexts are linked with this meta-topic, 3% of which come from topic 3, and more than 10% of the documents have at least one mention of this meta-topic (which is significant, when more than 50% of the documents are only associated with a single topic). While not a dominant across the whole dataset when compared to meta-topics such as "Geography" or "Medicine", it still consistently appears over the documents.

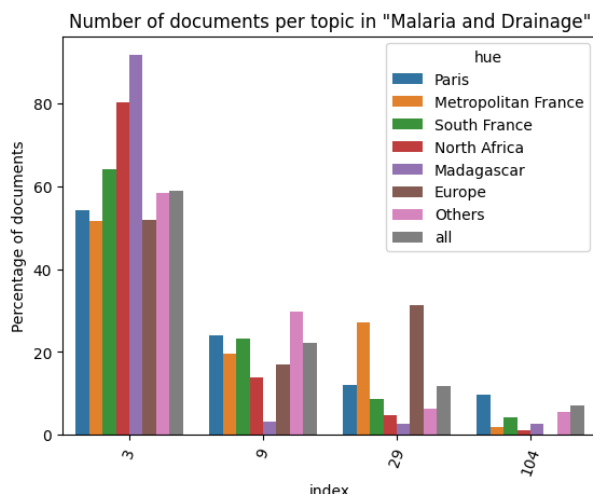
As a first start, we can see if the percentage of "Malaria" topics is similar across the different places of publication. Looking at the amount of documents that mention "Malaria and Drainage" per region, we find 6% as the lowest amount in South of France and 15% in Europe and Others, the rest being at the aforementioned 10%. The amount for the south of France is not that surprising, because as seen in Section 3.2, there is a high amount of documents linked with advertisements, which might skew the overall data. It is interesting however to note that the biggest amount comes from either Europe or "Others"⁹. To see if such differences come from specific topics, we can look at the distribution of the different "Malaria" topics, compared to their place of publication (Figure 3.18). Topic 3, the most "general" topic about the drainage process, with words such as "drainage", "absorbent" or "assainir", is heavily present in all regions. However, it is the most present in the regions most associated with the large plantations, such as Madagascar, North Africa (more than 20% compared to the overall corpus) and in a lesser degree in the south of France. When comparing with topic 9, which directly refers to the disease ("paludisme", "malaria", "miasmes"), we see a heavy drop in Madagascar and a lesser one in North Africa. As such, we could say that while eucalyptus were planted in the colonial regions in grand part to cleanse and terraform the marshes, the discourse there tends to avoid mentioning the disease and speak more about the terraforming aspects, while the discussion about the disease is done in the mainland, where most scientific literature is published from.

⁶Doughty, *The eucalyptus : a natural and commercial history of the gum tree*, p. 36

⁷Doughty, *The eucalyptus : a natural and commercial history of the gum tree*, p. 36

⁸Doughty, *The eucalyptus : a natural and commercial history of the gum tree*, p. 39

⁹which mostly includes other French colonies across the world, but that were too small in the corpus to be evaluated separately

**FIGURE 3.18**

Bar plot displaying the percentage of topics per place of publication, within the documents containing at least once the meta-topic "Malaria and Drainage".

Another interesting result from Figure 3.18 is the distributions of topic 29, which needs explaining. While at a first glance topic 29 would be linked with "Geography" instead of "Malaria and Drainage", with words such as "Rome", "campagne", "italien", it actually represents a specific policy of eucalyptus plantations in the Roman Campagna, to drain the marshes and purify the air, led by the Trappist abbey of Tre Fontaine¹⁰. Thus, it is not surprising to see that around 30% of documents published in Europe touch on this issue. However, metropolitan France also has around 25% of documents touching on this topic, which is surprising, as the rest of France does not. By looking over the documents, we see that 26% of all contexts in which topic 29 appears are in the book "Culture de l'eucalyptus à St-Paul-Trois-Fontaines (près Rome)", published in 1882 in Landerneau. As the title suggest, it speaks about the previously mentioned campaign of eucalyptus plantation around Rome, and the benefits of draining the marshes. While this monograph only does not explain the 10% difference between the general corpus and metropolitan France, it is a way to understand why this particular incident is talked about in the region, especially since the sample size is relatively small (only 36 documents in metropolitan France mention topic 29). Moreover, most documents in the region speaking about this topic are religious in nature, such as "La Semaine religieuse du diocèse de Rouen", which comes from the fact that it was clergymen that started these eucalyptus plantations, and thus their feat is relayed through the ecclesiastic press.

Then, we can see what other topics are associated with the meta-topic. By using the graph of co-occurrence, we can then see which topics appear the most next to the topics linked with drainage. Taking topic 3, as it is the biggest topic, it appears most frequently next to topic 7, 14 and 79. The first two are associated with botanical properties of eucalyptus. The first describes its physical properties such as height and circumference, while the second describes its temperature requirements. These appear often together (~200 documents), as either there is a need to study the plant before committing to these drainage projects, or to introduce what kind of tree is already being used in the new plantations to the public. Probably both are true, as we see that while 60% of these co-occurrence happen in scientific publications (monographs or scientific press), there are still 31% that appear in the generalist press. Lastly, we can look at topic 79, which is linked with the Mediterranean, notably the French Mediterranean coast. This shows where these projects are being established, again reinforcing the fact that it is in the Mediterranean basin that eucalyptus are massively being planted.

¹⁰'Eucalyptus Trees in the Roman Campagna.' In: *Hall's journal of health* 30.3 (Mar. 1883), p. 76

Finally, we can look at these topics over time (Figure 3.19). Mentions of drainage and malaria (topics 3 and 9) start appearing in 1870, with a peak in 1875-76. This time frame is linked with several key events, such as the general start of the implementation of large eucalyptus plantations or the end of the Algerian outbreak of the disease. With some decline both topics stay present throughout the years until the start of the war. More over, the curve of the two topics seems to follow similar trends, showing that they are often used together. As regarding topic 29, we do see an increase in 1875-76, followed by a progressive decline. This rise is probably linked with the first results of the Trappist's plantations, starting in 1869, with a gradual decrease, the further from the initial results.

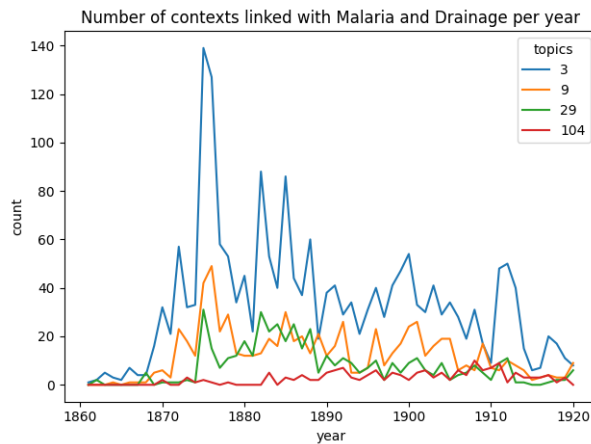


FIGURE 3.19

Line plot displaying the yearly number of contexts linked with topics 3, 9, 29 and 104.

In conclusion, while a key reason for the development of eucalyptus plantations, the topic of drainage and malaria is not the most discussed in the corpus. However, we do find that it is not insignificant and it is constantly found in all places of publication, with two main topics (3 and 9) in the biggest topics found in the corpus. This theme starts to appear around 1870, when the first large plantations used to sanitize marshes start to appear. Moreover, the model finds a specific event, with the plantation of eucalyptus by Trappist monks in the marshes of Rome. Being linked with the church, it explains why this Italian issue is talked in metropolitan France, though several church journals pushing the virtues of the plant. Moreover, the regular co-occurrence with topics describing the trees botanical characteristics show the way the drainage process is discussed, and which regions are part of the process.

4. DISCUSSION

Through this report, we have looked at the discourse around the eucalyptus tree. We have seen that there is a spike in the number of documents related to the tree around 1870, driven mostly by the generalist press, which cannot be fully explained by the general rise of published documents, thus showing a specific phenomenon happening at the time. We also discussed that these documents can be analyzed through a topic model, which, together with the plotting of documents per place of publication, reveals interesting key studies, such as Lyon or Madagascar. Moreover, it allows the analysis of specific themes, such as the topic of malaria. The latter which reveals that the topic of eucalyptus not only appears in the contexts where it might be expected (generalist press, monographs or official documents), but also through religious journals, which relay the work of eucalyptus plantations around Rome.

These results, while confirming intuitions about the discourse, still show some light on interesting ways it was shaped. Paris, being the cultural center of the francophone world is the city that really drives the subject. It is basically the only place that does publish about the plant before 1870 in the corpus, while still being largely ahead of the other regions after that date. Moreover, it is the center of any scientific production, publishing no less than 80% of all monographs of the corpus. However, we cannot neglect the contribution of the documents published in the regions of plantation, which also help shape the discourse locally. This is particularly striking in the topic distribution between the south of France and the colonies. By comparing the two case studies of Lyon and Madagascar, we see a clear difference in the topics mentioned as well as in the actors at play. While in the mainland it manifests mainly through advertisements, in Madagascar it is through the colonial institutions that eucalyptus appear in the documents (and the same phenomenon seems to occur in North Africa). Indeed, we see that the properties linked with the plant seem to not be used (or at least not advertised) in the colonies, while being heavily done so in the mainland. This can be seen through the disproportion between medicinal meta-topics (diseases, medicine and medication and chemical properties), which represent less than 5% of documents in the colonial regions, while often more than 15% in the mainland. We can thus easily see the domination of the mainland over its colonial empire. Paris and Lyon do use the plant not only for its drainage properties but also for its medicinal ones, while Madagascar and North Africa are only subjected to the former.

This conclusion can be linked with prior research on the tree. Eucalyptus as a colonial tool to "correct" the environment, so that it is better suited to the colonial interests, has been discussed before by Davin-Mortier¹, showing that it was used to sanitize marshes that could then be used, for example, for food cultivation or wood production. Such policies, marked by the high amount of institutional topics, do seem to be present in the regions presented here (North Africa, Madagascar). These studies being qualitative in nature, this project allows instead for a wider quantitative analysis of the subject, which does demonstrate a difference in the discourse between the colonizer and colonized regions.

Finally, we also have to discuss the numerous limitations of the study. The biggest one, as mentioned in Section 2.1.1 is related to the filtering of the files. Since Gallica's API limitations did not allow the download of all potential files, a filtering had to be put in place. However, the way it was implemented, while sounding relevant at the time, produced some biases related to which journals were or were not kept, as well as removing a large portion of the generalist press. The cracks show for example when examining

¹Davin-Mortier, *L'eucalyptus : un arbre aux ramifications environnementales, sociales et politiques en Méditerranée (XIXe-XXe siècles)*, pp. 3, 6

the biggest publishing cities in the south of France, where places such as "Toulouse", "Marseille" or "Bordeaux" do not show up in the top 5, whereas smaller towns do so. However, the regions results tend to show that a high amount of the discourse is used through advertisements, and it is probably the case for these cities as well. Nonetheless, a further study should take these places into account, so that a less biased corpus can be formed. Speaking of biases, we also need to take into account the biases deriving from the original Gallica database. While more abstract, the digitization process does involve some biases in which documents are digitized, how their metadata fields are filled out and more importantly which documents from the colonized regions are included. This might play a role in the amount of institutional topics in these regions, as the French libraries contributing to the database might have the more official and colonial documents, while not necessarily having access to or not having digitized the more local newspapers. This can also be linked with the language of the study, as these local newspapers would not all have been published in French, thus not entering this corpus. Then, as with any similar project, the OCR presents some limitations. The model used, not necessarily being suited for certain documents types such as the general press, makes it so that properly isolating the relevant context is a hard task to generalize. To mitigate this, we have chosen to take into account a very small context in order to take into account this issue. For a further analysis however, it would be recommended to not download Gallica's own text, but the page's scan and perform the OCR in-house, hopefully capturing better the structure of such documents. Finally, while performing a quantitative analysis of the discourse is useful to understand at a macro scale the dynamics and tendencies of a large corpus, it does not detract for the need of an in-depth qualitative analysis of the documents. While we tried in places to provide context directly from the documents, further studies should go more in-depth in this disrection. Notably, this analysis can provide the basis for discovering such documents that might have not been studied before.

5. CONCLUSION

In this project, we formed a corpus of documents related to the eucalyptus tree, which we then used to quantitatively analyze the discourse around the plant in the French-speaking world, between the European discovery of the tree and the end of the First World War. In this exploratory analysis, we tried to discover which actors shaped the discourse around the tree, and how they did it. Concerning the actors, the data shows several different types. We see that while at the beginning the discourse was enclosed within scientific publications, the press started taking control after 1870. While these documents are not as in-depth as the scientific monographs, they largely participate in raising general awareness about the plant. Moreover, by looking at places of publication, we see that there are different actors at play, between mainland France and its colonies. This can be seen in the way eucalyptus is talked about, between the cases of Lyon and Madagascar. The first driven by the generalist press and advertisements about the medicinal virtues of the plant, while the latter being used in the official state press, confined to institutional topics. This difference of treatment does portray the eucalyptus tree as a colonization tool. However, we do also find differences within mainland France, notably between the north, the south and Paris. While the south is mainly linked with adverts, we see that the north, where no plantations are found, the discourse is often slightly more scientific in nature (with most topics being linked to medicine, geography or botanic), with fewer generalist journals. Moreover, when generalist press appears, it seems to be linked with religious institutions, as seen with the Trappist's plantations in Roman Campagna. In Paris meanwhile, we find the overwhelmingly majority of monograph publication and thus the more scientific fields of medicine, while at the same time finding most of the press production of the country. It is, unsurprisingly, the cultural and scientific leader of the French-speaking world, and thus of the eucalyptus discourse.

However, while this quantitative approach works to understand who the actors are and how they tend to shape the discourse around eucalyptus, it struggles to understand their motivation. Indeed, this project only captures topics, without capturing the thought process behind certain political or editorial decisions, specific to each document. And while theories can be emitted, this approach is better suited as a complement to a proper qualitative research. In conclusion, this project contributes to the research on the discourse around eucalyptus, by showing the effects of certain policies in the general discourse, as well as finding new areas of potential research, while not having the ambition of replacing proper qualitative studies, key to better explaining the dynamics explored in this project.

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A. TOPICS

Word cloud for Topic -1
Medicine and Medication



Word cloud for Topic 1
Geography



Word cloud for Topic 3
Malaria and Drainage



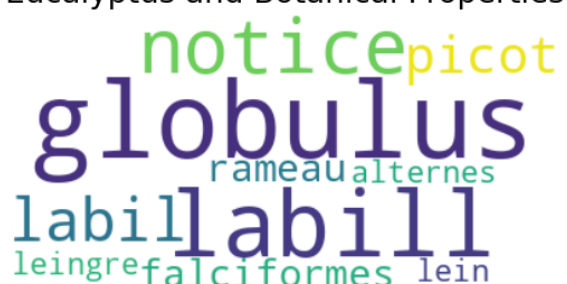
Word cloud for Topic 5
Errors



Word cloud for Topic 0
Geography



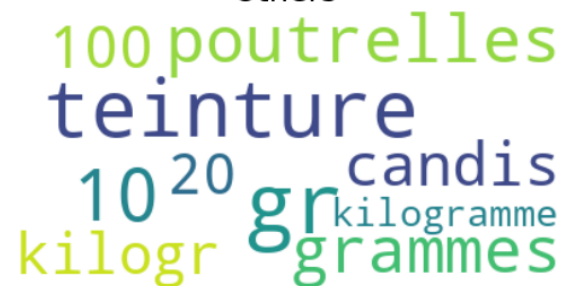
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Eucalyptus and Botanical Properties



Word cloud for Topic 4
Errors



Word cloud for Topic 6
Others



Word cloud for Topic 7
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Word cloud for Topic 9
Malaria and Drainage



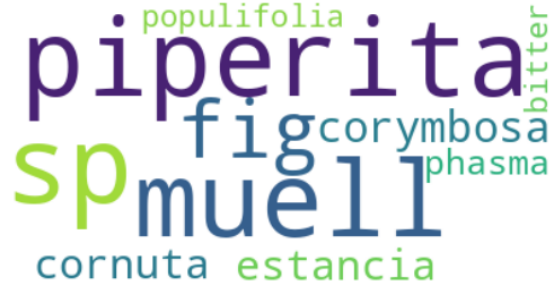
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Industry and Manufacturing



Word cloud for Topic 13
Descriptions



Word cloud for Topic 8
Fauna and Flora



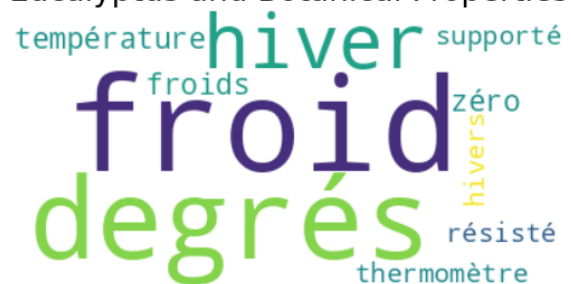
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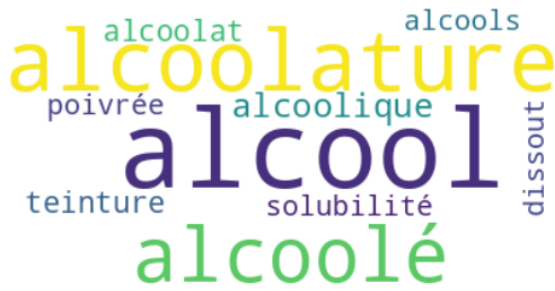
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Word cloud for Topic 14
Eucalyptus and Botanical Properties



Word cloud for Topic 15
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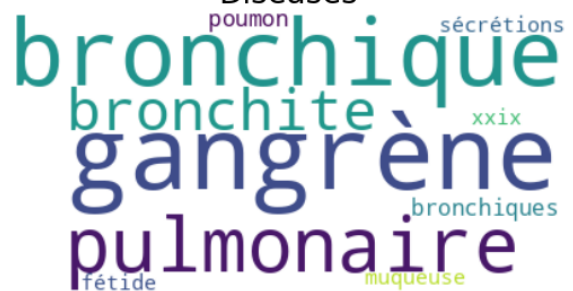
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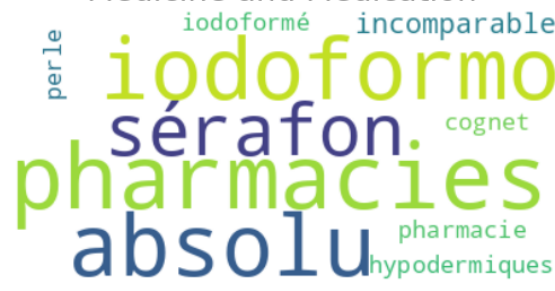
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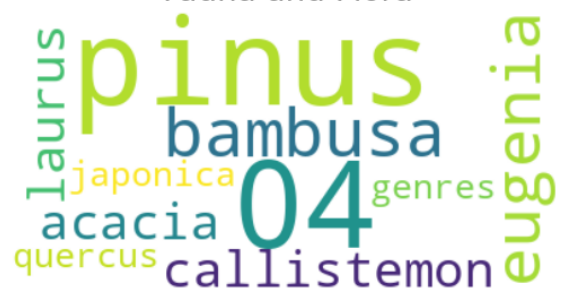
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Word cloud for Topic 19
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Word cloud for Topic 20
Fauna and Flora



Word cloud for Topic 21
Institutions and Transport



Word cloud for Topic 22
Chemical Properties



Word cloud for Topic 23
Medicine and Medication

hygiéniques phénique
tourbe
antiseptique
propriétés
antiseptiques
sanitaires
acide imputrescibles
scientifiques

Word cloud for Topic 25
Medicine and Medication

thérapeutiques
remède
médecins
thérapeutique
guérit remèdes
manifeste
fièvres médicales

Word cloud for Topic 27
Fauna and Flora

guilmeth eucalypté
noires miel ruche
naturel
hutte abeilles ruches

Word cloud for Topic 29
Malaria and Drainage

gildas romaine paul
italie rome
italien
campagne
fontaines abbaye trappistes

Word cloud for Topic 24
Eucalyptus and Botanical Properties

nombreuses appartiennent
espèces
noms caractères
connaît genre
colosses
espèce variétés

Word cloud for Topic 26
Chemical Properties

inconvenients
bichlor
cristallisé
représente
thérapeutique valeur eucalyptéol
hydrate
anthoine degré

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faits pivotante
réussi rien

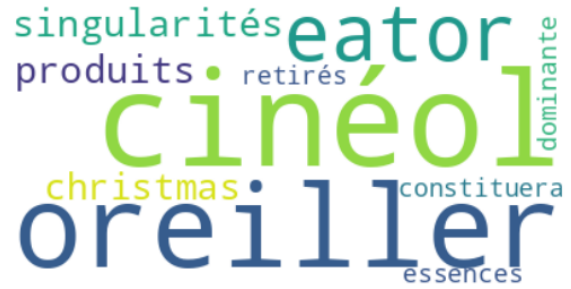
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odeur celle
aromatique parfum saveur

Word cloud for Topic 31
Medicine and Medication



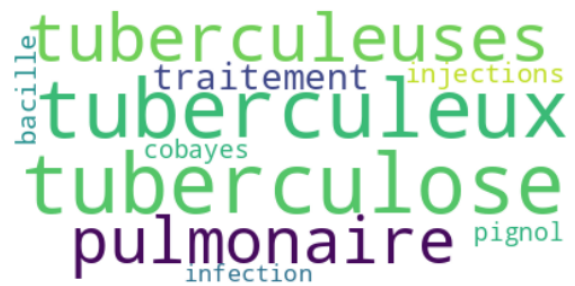
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Medicine and Medication



Word cloud for Topic 33
Descriptions



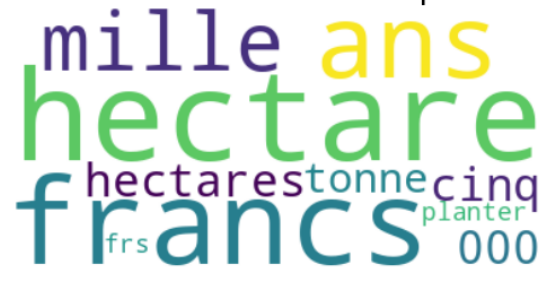
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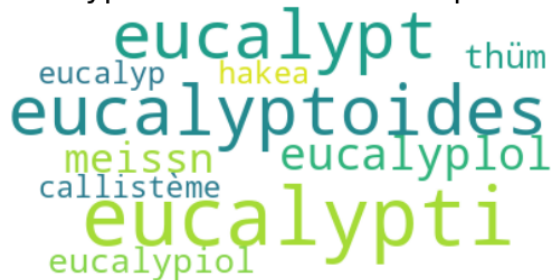
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Word cloud for Topic 36
Institutions and Transport



Word cloud for Topic 37
Eucalyptus and Botanical Properties



Word cloud for Topic 38
Others



Word cloud for Topic 39
Institutions and Transport

loi tabac interdit
régie
fiévet cigares
cigarettes
prospectus
fumer fumeurs

Word cloud for Topic 41
Others

mexicaines
rhumes
calyptus
seigle
viii
granulé
velpry
granules
ergoté

Word cloud for Topic 43
Geography

coutumes province
canton
fianarantsoa
malgaches dame
réquisition
née né district

Word cloud for Topic 45
Eucalyptus and Botanical Properties

ferd désinfect
massue cliniques
veucalyptus
filiformes naudiniana
globulus
globuleux labill

Word cloud for Topic 40
Geography

europa
laisseront
acclimater
introduction
midi
portugal
france
montpellier
espagne
acclimaté

Word cloud for Topic 42
Medicine and Medication

gommeuse centigrammes
jour
dose
prescrit journalière
bucquoy
grammes
potion
quotidienne

Word cloud for Topic 44
Diseases

créoline
variole
salol
solvéol
résorcine anthrax
sublimé
salicylé
rougeole accouchements

Word cloud for Topic 46
Others

vermifuge quina codex
guillet farine
kilo moutarde
elixir
litre pure

Word cloud for Topic 47
Others

bordée route rues conduit avenue bordent entourées roule kilomètres

Word cloud for Topic 49
Eucalyptus and Botanical Properties

graines ramel 1854 1856 1860 mueller apôtre hamel propagateur

Word cloud for Topic 51
Medicine and Medication

copahu cubèbe tablette ratanhia magnésie détacher ferrugineux chocolat benzine alun

Word cloud for Topic 53
Geography

nouvelle the hollandaise refouler défrichements plaisait perpétuellement originaires chassé abondaient

Word cloud for Topic 48
Medicine and Medication

gimbert thérapeutique gubler thérapeutiques leçons emploi globulus cannes professeur

Word cloud for Topic 50
Medicine and Medication

vierge dépôt savons paris gazeuses santé vinaigre meyerbeerbains dérivés

Word cloud for Topic 52
Medicine and Medication

injections cutanées hypodermiques souscutanées chateaubourg dater carbonique douloureuses traitement injection

Word cloud for Topic 54
Institutions and Transport

voie train chemin ferrée vide attendu palan freins passage prêt

Word cloud for Topic 55
Geography

voiry
algerie
trottier
algerie
in
25
role
brochure
icp

Word cloud for Topic 57
Medicine and Medication

bouillante
cuillerée
cuiller
désaltèrent
café
casserole
cuillerées
alcool
mélange
soupe

Word cloud for Topic 59
Medicine and Medication

toilette
dents
carie
préserve
dentifrice
guérissant
eucalypta
suez
maux
conserve

Word cloud for Topic 61
Chemical Properties

hydrocarbure
soluble
anhydre
phosphorique
hydrogène
phellandrène
chlorhydrique
cinéol
anhydride
cornue

Word cloud for Topic 56
Institutions and Transport

hectares
écuries
olivettes
remises
caves
sise
bâtiment
orangerie
palestro
territoire

Word cloud for Topic 58
Diseases

tuberculeuse
solutions
pleurésie
maladie
sûrement
amener
évolution
origine
arrêtent
gaiacol

Word cloud for Topic 60
Others

supérieur
eucalyptol
mission
niveau
beau
venue
hauteur
pleine
grande
bois

Word cloud for Topic 62
Chemical Properties

lepto
drag
lebrun
bmn
brun
ducœur
eucalyptine
maladencalyptine
brnn

Word cloud for Topic 63
Others

dépôt depot gal
séductions
cafés vente demander
comptoir
eucalypsinthe théories

Word cloud for Topic 65
Medicine and Medication

confiné ordonne
vapeurs sèche
résultent
vaporisations émétisante
inhalations
peintepulvérisations

Word cloud for Topic 67
Medicine and Medication

eucalyptéol forme véol
olivaire
olivâtre chaque
capsule
gr rougeoles contient

Word cloud for Topic 69
Others

enum obcordata
yearlings
resdoni preissiana risdoni
poulinières
tetraptera linkinfusé

Word cloud for Topic 64
Others

parents
mobilisés
informés
amis maison opéra
dérivés mobilises tél
meyerbeer

Word cloud for Topic 66
Others

costulata pratique
guide culture
usages
jelutong neish
théorique
james bananes

Word cloud for Topic 68
Fauna and Flora

deueucalyptus
ombelles pédoncule
fleurs
portées rameaux
valvaire fleur
générale eucalyptus
étamines

Word cloud for Topic 70
Others

représenter accordent
télégraphiques poteaux carvallo
renard bône
former trois
tristany

Word cloud for Topic 71
Institutions and Transport

poste assure
trains sidi 39
voiture correspondance
arrivant
partant roviggo

Word cloud for Topic 73
Medicine and Medication

mousnier douces
stérilisée
olive olives
huile anytol julep
gommeux macarthuri

Word cloud for Topic 75
Institutions and Transport

service conseil avis
pépinière
providence
filao plants
disposition crédit tient

Word cloud for Topic 77
Geography

regnans roiméan
francisco san
californien californ
californie
vauvert séquoia

Word cloud for Topic 72
Medicine and Medication

parachute savon
chalmin tonique
châtelaines
epidermal
thé pommade
vinaigre epidermale

Word cloud for Topic 74
Chemical Properties

berthelot exsudation
mélitose
mannesaccharose
sucree produite
sucree
trehalose
canne

Word cloud for Topic 76
Others

botanique huile
lehmanni
leucoxylon avril
concolor
né pot bai élégant

Word cloud for Topic 78
Others

guignard carnauba
admets
chateau antiseptiques
cod tkots
blancue
bonconsril
tinois

Word cloud for Topic 79
Geography

schisteux
méditerranée
méditerranéennes
marseille
méditerranéenne
méditerranéen
littoral
région
esterel
dégagement

Word cloud for Topic 81
Industry and Manufacturing

laine ordinaires
supérieurs
chers
flanelles
ouate
hygiéniques
vêtements
eucalyptus

Word cloud for Topic 83
Institutions and Transport

poste
matin
arrivant
voiture
sidi
assure
correspondance
trains
partant
réfaisant

Word cloud for Topic 85
Chemical Properties

obstrue
cannelle
pores
platine
aa
grphénol
thymol
mousse
étudia

Word cloud for Topic 80
Institutions and Transport

consistant
vouloir
déclaré
située
demandé
propriétaires
indivis
immatriculation
manioc
patates

Word cloud for Topic 82
Diseases

guéries
pourries
droguiers
agari
punctuation
stachetées
secrétrices
varioliqes
correspondent
cinés

Word cloud for Topic 84
Medicine and Medication

eucalyptoline
désinfectant
radicalement
automatique
contagion
germes
désinfecteur
odeurs
mauvaises
O
N
O

Word cloud for Topic 86
Others

boul
comptoir
arrêterai
caniche
austalet
auss
prés
vos
bjanc
iasman

Word cloud for Topic 87
Medicine and Medication



Word cloud for Topic 89
Chemical Properties



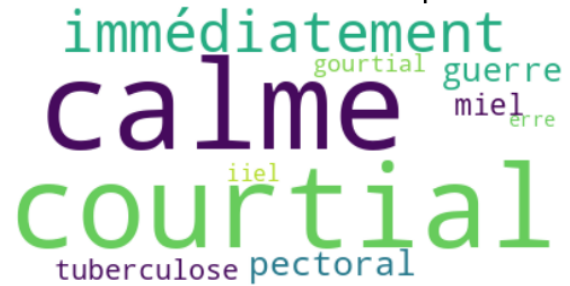
Word cloud for Topic 91
Geography



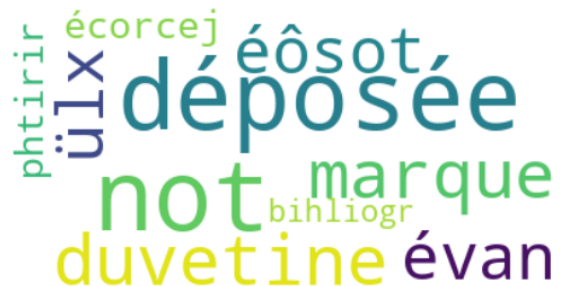
Word cloud for Topic 93
Eucalyptus and Botanical Properties



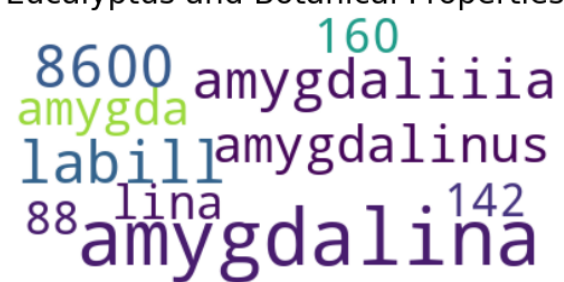
Word cloud for Topic 88
Institutions and Transport



Word cloud for Topic 90
Others



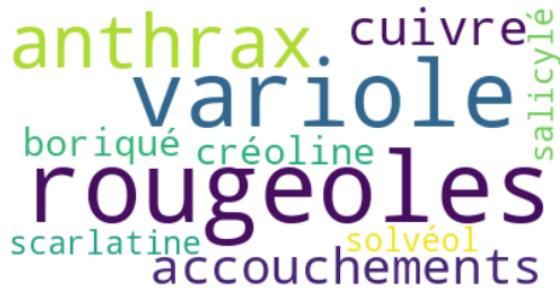
Word cloud for Topic 92
Eucalyptus and Botanical Properties



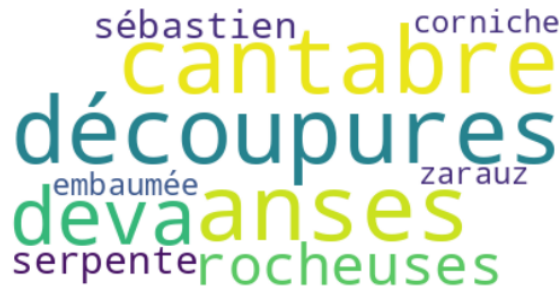
Word cloud for Topic 94
Errors



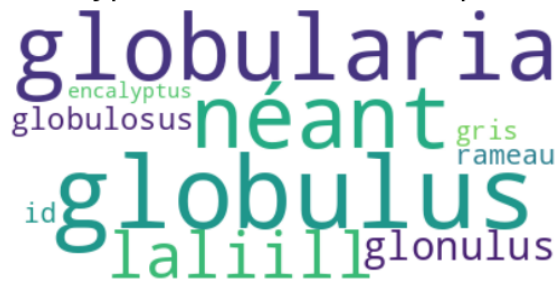
Word cloud for Topic 95
Diseases



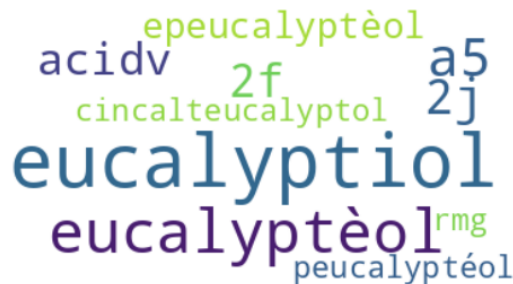
Word cloud for Topic 97
Others



Word cloud for Topic 99
Eucalyptus and Botanical Properties



Word cloud for Topic 101
Errors



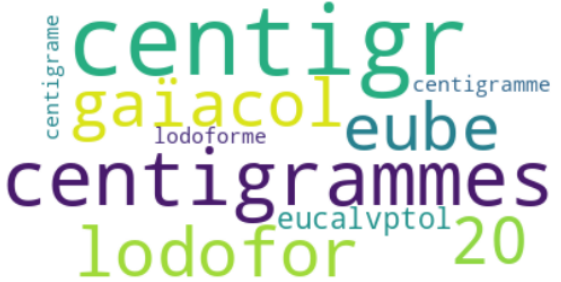
Word cloud for Topic 96
Geography



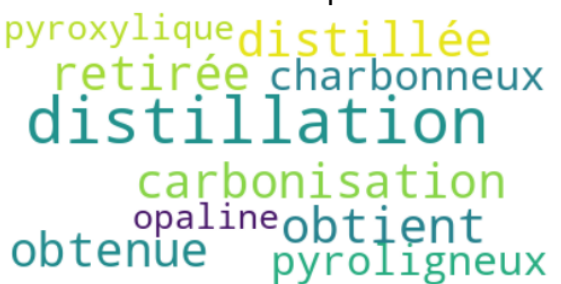
Word cloud for Topic 98
Industry and Manufacturing



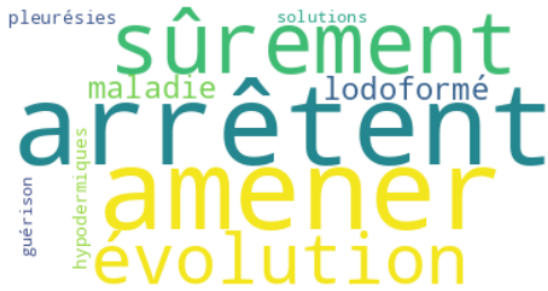
Word cloud for Topic 100
Chemical Properties



Word cloud for Topic 102
Chemical Properties



Word cloud for Topic 103
Diseases



Word cloud for Topic 105
Others



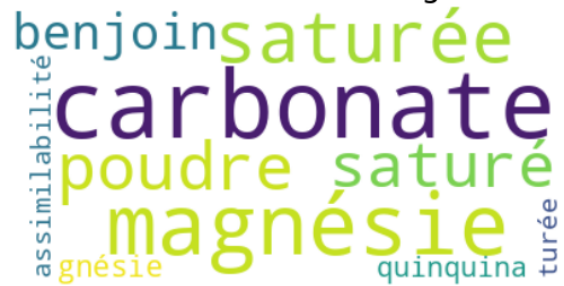
Word cloud for Topic 107
Medicine and Medication



Word cloud for Topic 109
Medicine and Medication



Word cloud for Topic 104
Malaria and Drainage



Word cloud for Topic 106
Errors



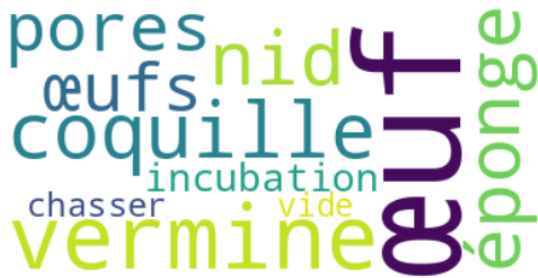
Word cloud for Topic 108
Fauna and Flora



Word cloud for Topic 110
Institutions and Transport



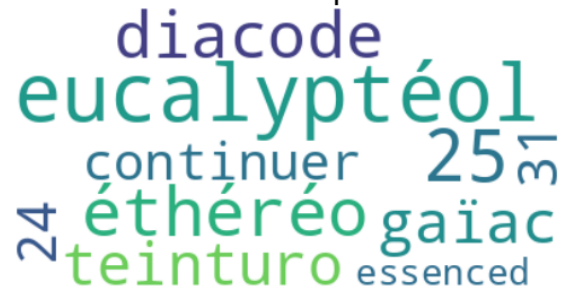
Word cloud for Topic 111
Fauna and Flora



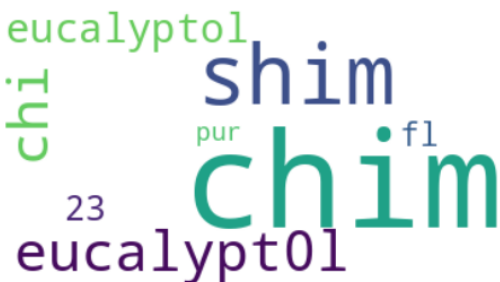
Word cloud for Topic 113
Medicine and Medication



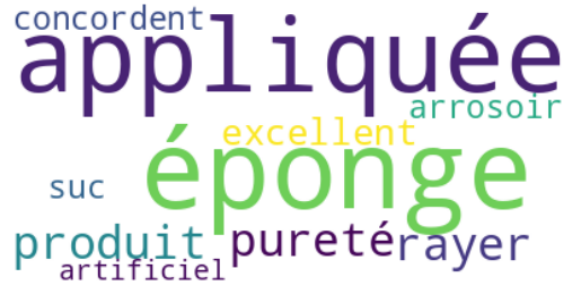
Word cloud for Topic 115
Chemical Properties



Word cloud for Topic 117
Errors



Word cloud for Topic 112
Industry and Manufacturing



Word cloud for Topic 114
Medicine and Medication



Word cloud for Topic 116
Diseases



Word cloud for Topic 118
Diseases



Word cloud for Topic 119
Descriptions

contourne accidentée
couronnée
détours
abrupte tantôt gorges
rocheuse traversons
exubérante

Word cloud for Topic 120
Diseases

capsule interne
bout repas
phtisie jours
pulmonaires premiers
lodoformés

Word cloud for Topic 121
Fauna and Flora

résorcinol tmctrasideras
resinifera
résinifera
backhousia bdy
sm suc
thiorésorcine smith

Word cloud for Topic 122
Others

sessiles longuement
opposées
alternes pétiolées
connées âge dirigées
adulte décussé

Word cloud for Topic 123
Errors

fjm méristèmes
cinéaueucalyptol a5
munier
1_87i acquoy jetm
javecvi f
lodoforme

Word cloud for Topic 124
Others

sublimé phéniqué
créolise
créoline
eucaly borlqué
boriqué
kucatypto créullue créoliae

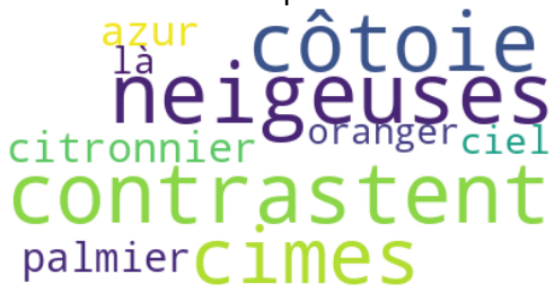
Word cloud for Topic 125
Errors

clicersicolor
tannin
actif lii
nommée eucalyp us
tol eucalypfus
centigr

Word cloud for Topic 126
Others

brûleraient
percées
chaudières asperger
enduire
proscrire engelures
recommande incrustation éviter

Word cloud for Topic 127
Descriptions



Word cloud for Topic 129
Others



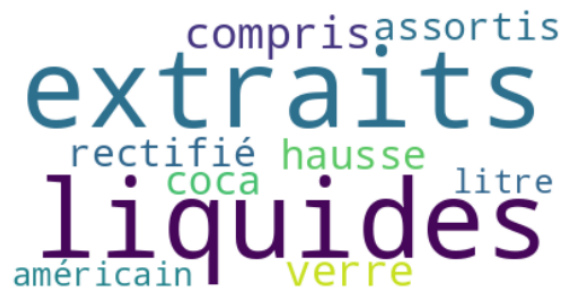
Word cloud for Topic 131
Errors



Word cloud for Topic 133
Descriptions



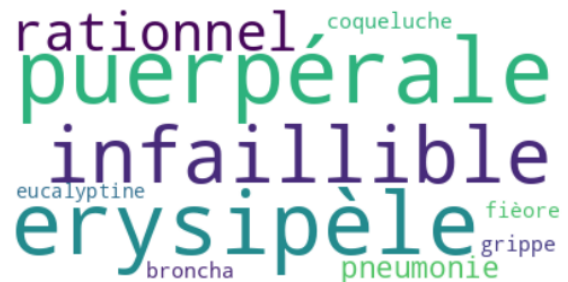
Word cloud for Topic 128
Others



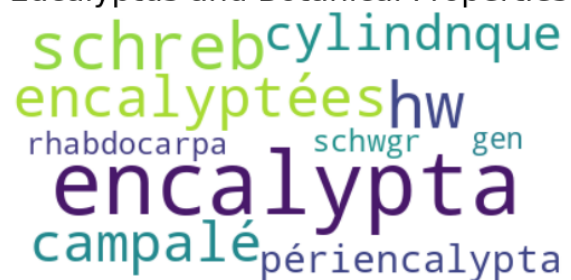
Word cloud for Topic 130
Institutions and Transport



Word cloud for Topic 132
Diseases



Word cloud for Topic 134
Eucalyptus and Botanical Properties



Word cloud for Topic 135
Industry and Manufacturing

fabrication
mascara
spéciale
ferry boufarik base
parfumerie
galtier hase exposaient